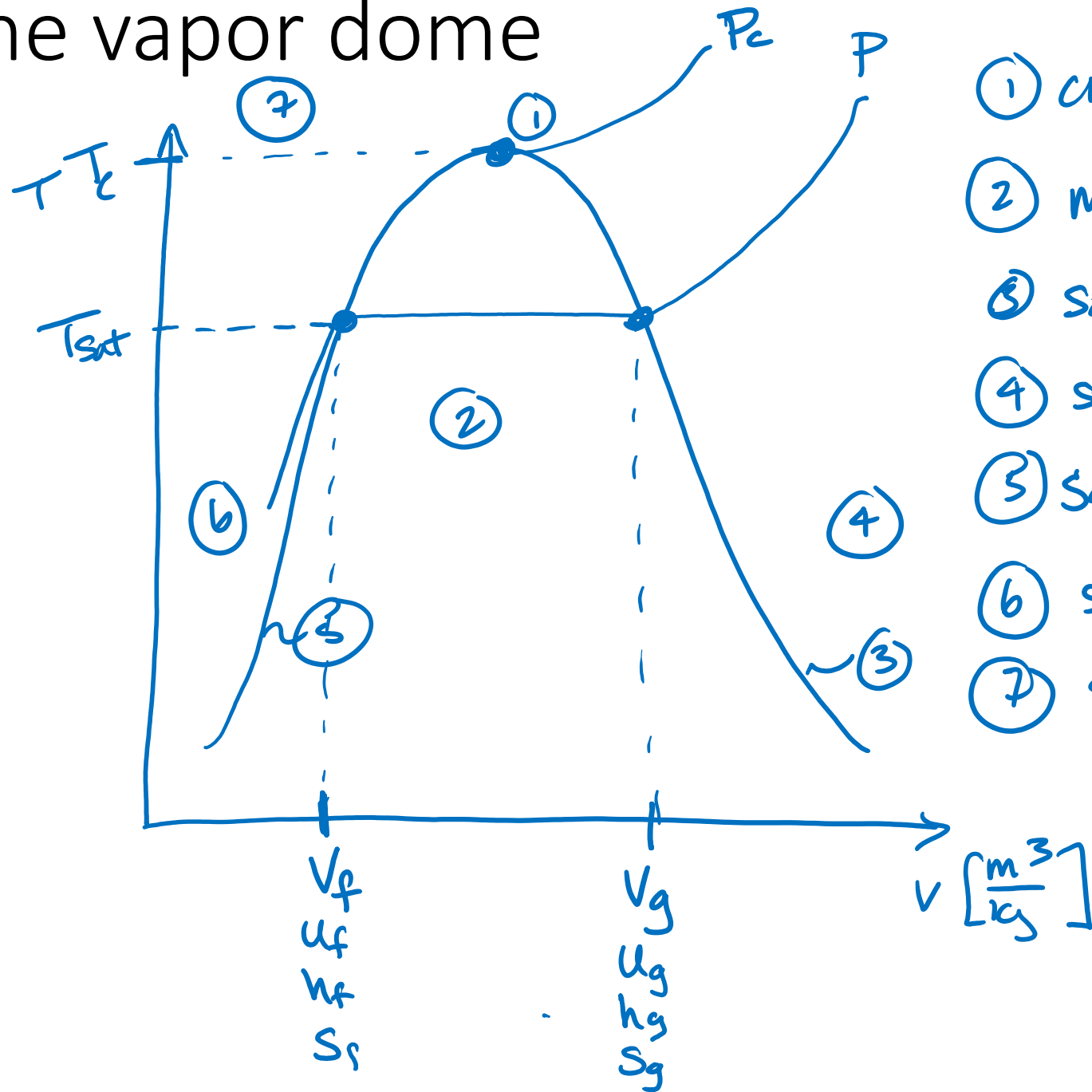




ME 300:
Engineering Thermodynamics

The Vapor Dome and Quality

The vapor dome



- ① critical point (T_c, P_c, v_c)
- ② mixture - liquid + vapor
- ③ saturated vapor
- ④ superheated vapor
- ⑤ saturated liquid
- ⑥ subcooled liquid
- ⑦ supercritical

Tables

$$x = \frac{M_g}{M_f + M_g}$$

$$x = \frac{V - V_f}{V_g - V_f} = \frac{u - u_f}{u_g - u_f} \dots$$

$$\begin{aligned} V &= x V_g + (1-x) V_f \\ &= x u_g + (1-x) u_f \\ &\vdots \end{aligned}$$

① Saturation tables
→ sat. vap
→ sat. liquid. } D.1 - T
D.2 - P

② Superheated vapor - D.3
→ supercritical table

③ Subcooled liquid - D.4

Example using the table

$$T = \underline{310 \text{ K}}$$

$$v = 22.903 \text{ m}^3/\text{kg}$$

$$P = ?$$

$$P_{\text{phase}} = ?$$

$$T_{\text{table}} = ?$$

1) Table D.1

$$\Rightarrow P_{\text{phase}} = \text{sat. vap}$$

$$2) P = P_{\text{sat}} = 6.231199 \text{ kPa}$$



What phase is this fluid?

- a) Saturated vapor
- b) Saturated liquid
- c) Mixture
- d) Vapor
- e) Liquid